# MATH 113: COMPLEX ANALYSIS

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## 1. Background

1.1. **The Complex Numbers.** Basic vector space structure. Euclidean metric. Definition of functions between complex numbers. Example: $f : z \to \frac{1}{z}$.

**Proposition 1.1** (W 1.7)**.** *Suppose* $D \subset \mathbb{R}^2$ *is open and connected and* $u : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ *is such that* $u_x \equiv 0 \equiv u_y$*. Then* $u$ *is constant.*

**Definition 1.2.** $F : D \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ *is differentiable at a interior point* $z_0 \in D$ *if there is a linear map* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *such that:*

$$\lim_{z \to 0} \frac{1}{||z||} ||F(z + z_0) - F(z_0) - T(z)|| = 0$$

**Remark.** *If* $F(z) = (u(x, y), v(x, y))$ *is differentiable at* $z_0$*, then* $u$ *and* $v$ *have first partials at* $z_0$ *and the matrix of* $T$ *with respect to the standard basis is:*

$$[T]_\sigma = \begin{pmatrix} u_x(z_0) & u_y(z_0) \\ v_x(z_0) & v_y(z_0) \end{pmatrix}$$

*The converse fails! W Ex. 1.2.2.*

**Theorem 1.3** (W 1.8)**.** *Suppose that* $F : D \subset \mathbb{R}^2 \to \mathbb{R}^2$ *is given by* $F(x, y) = (u(x, y), v(x, y))$ *and* $D_r(z_0) \subset D$*. If the first partials of* $u$ *and* $v$ *exist in* $D$ *and are continous at* $z_0$ *then* $F$ *is differentiable at* $z_0$*.*

If the partials are defined in the neighborhood of $z_0$, it makes sense to talk about their partials, giving us the 2nd derivatives, i.e. $u_{xx}, u_{xy}, u_{yx}, u_{yy}$.

**Theorem 1.4** (W 1.9, Clauraut)**.** *If* $u_{xy}$ *and* $u_{yx}$ *are continous at* $z_0$*, then* $u_{xy} = u_{yx}$*.*

Now we want to defined some interesting functions of complex variables.

**Definition 1.5.** *We define the complex exponential* $\exp : \mathbb{C} \to \mathbb{C}$ *by:*

$$\exp(x + iy) = e^x \cos(y) + ie^x \sin(y)$$

**Proposition 1.6** (W 1.16)**.** *The exponential has the following properties:*

*(1)* $e^{z+2\pi i} = e^z$.
*(2)* $e^{z+w} = e^z \cdot e^w$
*(3)* $e^{-1} = \frac{1}{e^z}$
*(4)* $e^z = e^w \iff w = z + 2\pi i k$ *for* $k \in \mathbb{Z}$.

*Proof.* Problem 1 on PSet 1. $\square$

**Definition 1.7.** *We say $\tau \in \mathbb{C}$ is unimodular if $|z| = 1$. This happens if and only if $z = e^{i\theta}$ when $\theta \in \mathbb{R}$.*
*If $z \in \mathbb{C} \setminus \{0\}$, then the polar form is:*

$$z = |z|\frac{z}{|z|} = re^{i\theta}$$

*where $\theta$ is called the argument. We have:*

$$\arg(z) = \{\theta \in \mathbb{R} : z = |z|e^{i\theta}\}$$

*and if $z = re^{i\theta_0}$ then:*

$$\arg(z) = \{\theta_0 + 2\pi k : k \in \mathbb{Z}\}$$

*Note that* arg *is a set-valued function in $\mathbb{C} \setminus \{0\}$.*

**Proposition 1.8** (W 1.2.2)**.** *Let $f(z) = \exp(z)$. Then, $f$ maps $\mathbb{C}$ to $\mathbb{C} \setminus \{0\}$. Then,*

$$\{w \in \mathbb{C} : e^w = z\} = \{\ln(|z|) + i\theta : \theta \in \arg(z)\} = \ln(|z|) + i\arg(z)$$

**Definition 1.9.** *If $z \neq 0$, then we define the (set-valued) logarithm:*

$$\log(z) = \ln(|z|) + i\arg(z)$$

Now, we need to talk about picking branches to get honest-to-goodness functions.

**Definition 1.10.** *Suppose that $f$ is a set-valued function on a set $D$. Then a branch of $f$ is $D$ is a continous function $F : D \to \mathbb{C}$ such that:*

$$F(z) \in f(z) \quad \forall z \in \mathbb{C}$$

**Example.** *Let $z \in \mathbb{C} \setminus \{0\}$ then $\exists! \theta \in (-\pi, \pi]$ such that $\theta \in \arg(z)$. Then we can define:*

$$\mathrm{Arg} : \mathbb{C} \setminus \{0\} \to (-\pi, \pi]$$

*such that $\mathrm{Arg}(z) \in \mathrm{arg}(z)$.*
*This has a jump discontinuity on $(-\infty, 0]$ but is continous elsewhere. See W Lemma 1.26.*

A branch of arg in $D$ gives us a branch of the logarithm in $D$. We call:

$$\mathrm{Log}(z) = \ln(|z|) + i\,\mathrm{Arg}(z)$$

the principal branch of log in $D^*$.

This is highly non-unique. One thing we would want to show is that there is no cut you can make to get a logarithm continous everywhere.

**Exercise 1.4.1.** Show the following properties for the complex exponential:

(1) $e^{z+2\pi i} = e^z$
(2) $e^{z+w} = e^z e^w$
(3) $e^{z-w} = \frac{e^z}{e^w}$
(4) $e^{-z} = \frac{1}{e^z}$

**Exercise 1.4.8(a).** Use (1.6), De Moivre's Formula, and the geometric sum $1+z+\cdots z^n = \frac{z^{n+1}-1}{z-1}$ to prove the following assuming $0 < \theta < 2\pi$.

$$1 + \cos(\theta) + \cdots + \cos(n\theta) = \frac{1}{2} + \frac{\sin((n+1/2)\theta)}{2\sin(\theta/2)}$$

**Exercise 1.4.9.** Observe that the $n$th roots of $1$ are given by $\{1, \omega, \cdots, \omega^{n-1}\}$, where $\omega = e^{2\pi i/n}$. Assuming $n > 1$, show that $1 + \omega + \cdots + \omega^{n-1} = 0$. Prove a similar result for the $n$th-roots of $-1$.

**Exercise 1.4.13.** Suppose that $p(z) = a_n z^n + \cdots + a_1 z + a_0$ has real coefficients.

(1) Show that if $z_0$ is a root of $p$, then either $z_0$ is real or $\overline{z}_0$ is also a root of $p$.
(2) Assuming the Fundamental Theorem of Algebra, show that $p$ must factor as a product:
$$p(z) = a_n(z - r_1)\cdots(z - r_k)q_1(z)\cdots q_s(z)$$
(3) Factor $p(z) = z^4 + z^3 + z^2 + z + 1$ into a product of irreducible quadratics.

1.2. **Sequences and Series.** Recall that if $(a_n)_{n=1}^{\infty}$ is a sequence in $\mathbb{C}$ then the series

$$\sum_{n=1}^{\infty} a_n$$

converges if the sequence $(s_n)$ of partial sums converges and then:

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n$$

We say the series converges absolutely if $\sum_{n=1}^{\infty} |a_n| < \infty$.

**Lemma 1.11** (W 1.2.9)**.** *An absolutely convergent series is convergent.*

*Proof.* Suppose $\sum_{n=1}^{\infty} |a_n| < \infty$ is absolutely convergent. Then given $\epsilon > 0$, $\exists N$ such that $n \geq N \implies \sum_{n=N}^{\infty} |a_n| < \epsilon$. Now if $m > n \geq N$, we have:

$$|s_m - s_n| \leq \sum_{k=n+1}^{m} |a_k| \leq \sum_{k=N}^{\infty} |a_k| < \epsilon$$

Thus, $(s_n)$ is Cauchy and by completeness of $\mathbb{C}$, it converges. $\square$

**Theorem 1.12** (W 1.1.30)**.** *Let* $(a_n) \subset \mathbb{C}$*. Then,*

*(1) If* $|a_n| < b_n$ *and* $\sum_{n=1}^{\infty} b_n < \infty$*, then the series* $\sum a_n$ *is absolutely convergent.*
*(2) If* $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$*, then* $\sum_{n=1}^{\infty} a_n$ *is absolutely convergent if* $L < 1$ *and diverges if* $L > 1$*.*
*(3) Let* $L = \limsup |a_n|^{1/n}$*. If* $L < 1$*, then* $\sum a_n$ *converges absolutely and diverges if* $L > 1$*.*

Recall, the definition of a limsup. For $(a_n)$, let $\alpha_n = \sup\{a_k : k \geq n\}$. This is monotonic decreasing. Then, $\limsup_n a_n = \lim_n \sup_{k\geq n} a_k$.

Recall, if $(f_n)$ is a sequence of $\mathbb{C}$ valued functions on $D \subseteq \mathbb{C}$ then we say $(f_n)$ converges pointwise on $D$ to some function $f : D \subset \mathbb{C} \to \mathbb{C}$ if:

$$\lim_{n\to\infty} f_n(z) = f(z) \qquad \forall z \in D$$

We say $(f_n)$ converges uniformly to $f$ on $D$ if $\forall \epsilon > 0$, there exists $N$ such that $n \geq N$ implies

$$|f_n(z) - f(z)| < \epsilon \qquad \forall z \in D$$

We use the same terminology for the series $\sum_{n=1}^{\infty} f_n$ by condering the appropriate partial sums.

**Theorem 1.13** (W 1.31, Weierstrass M-test)**.** *Suppose* $D \subseteq \mathbb{C}$ *and* $(f_n)$ *is a sequence of* $\mathbb{C}$*-valued functions on* $D$ *such that* $|f_n(z)| \leq M_n$ *for all* $z \in D$*. If* $\sum_{n=1}^{\infty} M_n < \infty$*, then* $\sum_{n=1}^{\infty} f_n$ *converges uniformly on* $D$ *to some* $f : D \subseteq \mathbb{C} \to \mathbb{C}$*.*

*Proof.* Note that $\forall z \in D$, $\sum_{n=1}^{\infty} f_n(z)$ converges to some $f(z) \in \mathbb{C}$ by comparision test. If $\epsilon > 0$, then $\exists N$ such that $n \geq N \implies \sum_{n=N}^{\infty} M_n < \epsilon$. If $n \geq N$, then

$$\begin{aligned} |f(z) - s_n(z)| &= |f(z) - \sum_{k=1}^{n} f_k(z)| \\ &= \left| \sum_{k=n+1}^{\infty} f_k(z) \right| \\ &\leq \sum_{k=n+1}^{\infty} |f_k(z)| \\ &\leq \sum_{k=n+1}^{\infty} M_k < \epsilon \end{aligned}$$

□

**Definition 1.14.** *Let* $(a_n)_{n=0}^{\infty} \subset \mathbb{C}$ *and* $z_0 \in \mathbb{C}$*, then the series:*

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1(z - z_0) + \cdots$$

*is called a power series centered at* $z_0$ *with coefficients* $a_k$*.*

**Theorem 1.15** (W 1.34)**.** *Consider*

$$P : \sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1(z - z_0) + \cdots$$

*Then, there exists* $R \in [0, \infty]$ *such that:*

*(1)* $P$ *converges absolutely if* $|z - z_0| < R$
*(2)* $P$ *diverges if* $|z - z_0| > R$
*(3) and if* $0 < r < R$*, then* $P$ *converges uniformly on* $D_r(z_0)$*.*

*Furthermore,*

$$\frac{1}{R} = \text{limsup}_n |a_n|^{1/n}$$

*Proof.* Can assume $z_0 = 0$ without loss of generality. $P$ always converges at $z = 0$. If $P$ converges only at $z = 0$, then take $R = 0$ so that $(a)$ and $(b)$ hold and $(c)$ is vacuous.

Otherwise, suppose the $P$ converges at another point $z_1$ with $|z_1| > 0$. Then, $\exists M$ such that:

$$|a_n z_1^n| \leq M \qquad \forall n$$

If $|z| < |z_1|$, then,

$$|a_n z^n| = |a_n z_1^n| \left| \frac{z}{z_1}^n \right| \leq M \left| \frac{z}{z_1}^n \right|$$

and $P$ converges absolutely at $z$ by the comparison test.

Let $R = \sup\{|z| : P \text{ converges at } z\}$. If $|z| < R$, then $\exists z$ such that $P$ converges at $z$ and $|z| < |z_1|$. But then, $P$ converges absolutely at $z$. If $|z| > R$, we know that $P$ diverges at $z$. Thus, $(a)$ and $(b)$ hold with $R$.

If $0 < r < R$, then $P$ converges absolutely at $z = r$, and we can let $M_n = |a_n| r^n$ and then $P$ converges uniformly on $D_r(0)$ and (c) holds. Now let,

$$R_0^{-1} = \limsup_n |a_n|^{1/n}$$

If $0 < |z| < R_0$, then

$$\limsup_n |a_n z^n|^{1/n} = R_0^{-1}|z| < 1$$

Thus, $P$ converges absolutely at $z$.

If $|z| > R_0$, then,

$$\limsup_n |a_n z^n|^{1/n} = R_0^{-1}|z| > 1$$

Thus, $P$ diverges.

Thus, $R_0 = R$. $\square$

Recall, that a subset $D \subset \mathbb{C}$ is connected if it does not admit a non-trivial partition into relatively open subsets.

**Remark.** *$A \subset \mathbb{C}$ is connected if given open sets $U, V \in \mathbb{C}$ such that $\{A \cap U, A \cap V\}$ is a partition and $A \cap U$ is not empty, then $A \cap V$ is empty.*

**Definition 1.16.** *We call a connected, open subset $D \subset \mathbb{C}$ a domain.*

If $E \subset \mathbb{C}$ and $z \in E$, then

$$E(z) := \bigcup \{A \subset \mathbb{C} : z \in A \subset E \text{ and } A \text{ connected}\}$$

Then $E(z)$ is connected and if $E(z) \cap E(w) \neq 0$ then, $E(z) = E(w)$. These form a partition of $A$ and are called the connected components of $E$.

Since disks $D_r(z_0)$ are connected, it follows that the connected components of an open set $U$ are open. (W Ex. 2.1.2). That is, every open subset is a countable union of domains.

We say $E \subset \mathbb{C}$ is path connected if given $z, w \in E$, there is a continous function:

$$\gamma : [0, 1] \to E$$

such that $\gamma(0) = z$ and $\gamma(1) = w$. Since $[0, 1]$ is connected, it follows that path-connected sets are connected.

For an connected open set, the converse holds (because disks are path-connected).

Questions for next time. $(a_n) \subset \mathbb{R}$. Suppose $\sum_{n=1}^{\infty} a_n$ converges.

Q1. Does $\sum_{n=1}^{\infty} a_n^2$ converge?
Q2. Does $\sum_{n=1}^{\infty} a_n^3$ converge?

## 2. Complex Differentiability

**Definition 2.1.** *Suppose that* $f : D \subset \mathbb{C} \to \mathbb{C}$ *and that* $z_0 \in D$ *is an interior point. Then* $f$ *is complex differentiable at* $z_0$ *if*

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

*exists. Then we call the* $f'(z_0)$ *the derivative of* $f$ *at* $z_0$*. If* $D$ *is open and* $f'(z)$ *exists for all* $z \in D$*, then we say that* $f$ *is holomorphic on* $D$ *and write* $H(D)$ *for the set of all holomorphic functions on* $D$*. If* $f \in H(\mathbb{C})$*, then we say* $f$ *is entire.*

**Remark.** *If* $f(z) = z^n$ *for* $n \in \mathbb{N}$*, then* $f'(z) = nz^{n-1}$ *with the usual proof. Similarly, the sum, product, quotient, and chain rules all apply with the usual proofs.*

**Remark.** *Polynomials are entire. Rational functions are holomorphic on their natural domains.*

**Example.** *Let* $f(z) = |z|^2$*. Thus, if* $z = x + iy$*,* $f(z) = x^2 + y^2$*. Consider* $\mathbb{C} = \mathbb{R}^2$*, this function is* $(x, y) \mapsto (x^2 + y^2, 0)$*. Recall, that* $|z|^2 = z\bar{z}$*. Thus,*

$$\begin{aligned} f'(z) &= \lim_{h \to 0} \frac{|z + h|^2 - |z|^2}{h} \\ &= \lim_{h \to 0} \frac{z\bar{h} + \bar{z}h + h\bar{h}}{h} \\ &= \bar{z} + z \lim_{h \to 0} \frac{\bar{h}}{h} \end{aligned}$$

*Suppose* $f'(z)$ *exists and* $h = x + iy$*. Then,*

$$\begin{aligned} f'(x) &= \bar{z} + z \lim_{x+iy \to 0} \frac{x + iy}{x + iy} \\ &= \bar{z} + z \lim_{x \to 0} \frac{x}{x} = \bar{z} + z \end{aligned}$$

*Simultaneously,*

$$f'(z) = \bar{z} + z \lim_{y \to 0} \frac{-y}{y} = \bar{z} - z$$

*These force* $z = 0$ *and* $f'(0) = 0$*.*

**Theorem 2.2** (W 2.4)**.** *Suppose* $r > 0$ *and*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

*for* $z \in D_r(z_0)$*. Then* $f \in H(D_r(z_0))$ *and*

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$$

*Proof.* Since $\lim_{n \to \infty} n^{1/n} = 1$, the radius of convergence of the function and its derivative are equal. Thus, $\forall z \in D_r(z_0)$ it suffices to see that $f'(z)$ exists and equals $g(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$.

Using a change of variables, we can assume $z_0 = 0$. Fix $z \in D_r(0)$ and $p$ such that $0 \leq |z| < p < r$. If $w \neq z$, then,

$$\begin{aligned}\frac{f(w) - f(z)}{w - z} - g(z) &= \sum_{n=1}^{\infty} \left( \frac{a_n w^n - a_n z^n}{w - z} - n a_n z^{n-1} \right) \\ &= \sum_{n=1}^{\infty} a_n A_n\end{aligned}$$

where $A_1 = 0$ and for $n \geq 2$,

$$\begin{aligned}A_n &= \frac{w^n - z^n}{w - z} - n z^{n-1} \\ &= \sum_{k=0}^{n-1} w^{n-1-k} z^k - n z^{n-1}\end{aligned}$$

We claim this equals,

$$(w - z) \sum_{k=1}^{n-1} k z^{k-1} w^{n-k-1}$$

To see this, expand it out.

Thus if $|w| < p$ and $n \geq 2$, then

$$\begin{aligned}|A_n| &\leq |w - z| \sum_{k=1}^{n-1} k p^{n-2} \\ &= |w - z| p^{n-2} \frac{n(n-1)}{2}\end{aligned}$$

Thus if $|w| < p$,

$$\left| \frac{f(w) - f(z)}{w - z} - g(z) \right| \leq |w - z| \sum_{n=2}^{\infty} n^2 |a_n| p^{n-2}$$

But, we also have

$$\limsup_n |a_n|^{1/n} = \frac{1}{R} \qquad p < r \leq R$$

This implies,

$$\sum_{n=2}^{\infty} n^2 |a_n| p^{n-2}$$

converges to some $M > 0$.

Therefore, $|w| < p$, then

$$\left| \frac{f(w) - f(z)}{w - z} - g(z) \right| \leq |w - z| M$$

Thus $f'(z)$ exists and equals $g(z)$. $\square$

**Corollary 2.2.1** (W 2.7)**.** *If* $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$ *for all* $z \in D_r(z_0)$*, then* $f$ *has derivatives of all orders in* $D_r(z_0)$ *and the* $a_n = \frac{f^{(n)}(z_0)}{n!}$*. In particular, the* $a_n$ *are uniquely determined by* $f$ *at* $z_0$*.*

*Proof.* Not difficult. Induction. □

Look at problem 2.1.8 for a similar result on Laurent series.

**Definition 2.3.** *If* $f$ *has derivatives of all orders at* $z_0$*, then*

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

*is the taylor series for* $f$ *about* $z_0$*.*

This corollary is very important. In the real case, such a taylor series may not converge or may converge to something besides $f$. **TODO:** Look into this.

As a consequence, take,

$$\frac{1}{1-z} = 1 + z + z^2 + \cdots$$

Hence,

$$\frac{1}{1-z^2} = 1 - z^2 + z^4 - z^6$$

You can tell all the derivatives of $\frac{1}{1-z^2}$ from these coefficients.

**Definition 2.4.** *Suppose that* $D \subset \mathbb{C}$ *is open. We say* $f : D \subset \mathbb{C} \to \mathbb{C}$ *is analytic if it is locally given by a power series. That is,* $\forall z_0 \in D$*, there exists* $r > 0$ *and* $a_n$ *such that* $D_r(z_0) \subset D$ *and*

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^r \qquad \forall z \in D_r(z_0)$$

**Remark.** *Clearly, analytic implies holomorphic. But analytic is less easy than this seems. Currently, we have written power series centered at some point* $z_0$*. This requires that the functions can be written as a power series centered at any point* $z \in D$ *in some open disk around* $z$ *contained in* $D$*.*

Note: Exercise 2.17.

Now we give an alternative characterization of holomorphic. Given $f : D \subset \mathbb{C} \to \mathbb{C}$,

$$f(x+iy) = u(x,y) + iv(x,y)$$

where $u, v : D \subset \mathbb{R}^2 \to \mathbb{R}$. The companion function is $F(x,y) = (u(x,y), v(x,y))$.

**Example.** *For instance, take*

$$f(x+iy) = (x+iy)^3 = x^3 - 3xy^2 + i(3x^2y - y^3)$$

*So* $u = x^3 - 3xy^2$ *and* $v = 3x^2y - y^3$*.*

**Definition 2.5.** *We say* $f(x+iy) = u(x,y)+iv(x,y)$ *satisfies the Cauchy-Riemann equations at* $z_0$ *if*

$$u_x(z_0) = v_y(z_0) \qquad u_y(z_0) = -v_x(z_0)$$

*Note that,*

$$f_x(z_0) = \lim_{\substack{x\to 0\\ x\in\mathbb{R}}} \frac{f(z_0+x)-f(z_0)}{x} = u_x(z_0) + iv_x(z_0)$$

*and similarly for* $f_y$*. Using this notation, these become*

$$f_x(z_0) = -if_y(z_0)$$

**Theorem 2.6.** *If* $f(x+iy) = u(x,y)+iv(x,y)$ *and* $F$ *is the companion function. Assume* $z_0$ *is an interior point of* $D$*. Then if* $f$ *is complex differentiable at* $z_0$ *if and only if* $F$ *is differentiable at* $z_0$ *and the Cauchy-Riemann equations hold at* $z_0$*. In the case* $f'(z_0)$ *exists, then,*

$$f'(z_0) = f_x(z_0) = -if_y(z_0)$$

*Proof.* Suppose $f'(z_0)$ does exist. Then, by definition,

$$\begin{aligned} f'(z_0) &= \lim_{x+iy\to 0} \frac{f(z_0)+x+iy-f(z_0)}{x+iy} \\ &= \lim_{x\to 0} \frac{f(z_0+x)}{x} = f_x(z_0) = f_x(z_0) \\ &= \lim_{y\to 0} \frac{f(z_0+iy)-f(z_0)}{iy} = -if_y(z_0) \end{aligned}$$

The rest is Exercise 2.2.5 on the homework. □

**Example.** *Consider* $f(z) = \bar{z}$*. Then,* $u(x,y) = x$ *and* $v(x,y) = -y$*. Then* $u_x \equiv 1 \not\equiv -1 \equiv v_y$*. Thus, this function is not complex differentiable anywhere. But* $F(x,y) = (x,-y)$ *is pretty nice!*

**Theorem 2.7** (W 2.15)**.** *Suppose* $D$ *is open and* $F : D \subset \mathbb{C} \to \mathbb{C}$ *has continuous first partials in* $D$*. If Cauchy-Riemann equations hold throughout* $D$*, then* $f \in H(D)$*.*

*Proof.* We have to show that if you have continuous partials, then the companion function is differentiable. □

**Corollary 2.7.1.** *The complex exponential function* $f(z) = \exp(z)$ *is entire and* $f'(z) = f(z)$ *for all* $z \in \mathbb{C}$*.*

*Proof.* We have $f(x+iy) = e^x\cos(y) + ie^x\sin(y)$. Thus,

$$\begin{aligned} u_x(x,y) &= e^x\cos(y) = v_y \\ u_y(x,y) &= -e^x\sin(y) = -v_x \end{aligned}$$

Thus, the CR equations hold everywhere, i.e. $f$ is entire. Moreover,

$$f'(x+iy) = f_x(x+iy) = e^x\cos(y) + ie^x\sin(y) = f(x+iy)$$

□

**Remark.** *If* $f(z) = \exp(z)$ *then* $f$ *has derivatives of all orders. Thus, it has a Taylor series at* $z_0 = 0$*,*

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

**Remark.** *A taylor series, generally, does not have to have anything with the original function. It could not converge outside of the center, it can converge but not agree with the original function except at the center, etc, in the real case. This won't happen to us in the complex world.*

**Proposition 2.8.** *Suppose* $D$ *is a domain and* $f \in H(D)$*. If* $f'(z) = 0$ *for all* $z \in D$*, then* $z$ *is constant.*

*Proof.* Let $f(z) = u(z) + iv(z)$. Since $f'(z) = f_x(z) = -if_y(z)$, this means,

$$u_x \equiv u_y \equiv v_x \equiv v_y \equiv 0$$

As the domain is connected, $u$ and $v$ are constant, hence $f$ is constant. □

**Lemma 2.9** (2.19)**.** *Suppose* $f$ *is a branch of* $\log(z)$ *on a domain* $D$*. Then* $\Re(f(z)) = \ln(|z|)$*,* $f \in H(D)$*, and* $f'(z) = \frac{1}{z}$ *for all* $z \in D$*.*

*Proof.* By definition, $f$ is continuous and $\exp(f(z)) = z$ for all $\forall z \in D$. Taking modulus on both sides, we get,

$$\begin{aligned} |z| = |\exp(f(z))| &= \exp(\Re(f(z))) \\ \implies \ln(|z|) &= \Re(f(z)) \end{aligned}$$

Now let $z_0 \in D$. We want to show,

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \frac{1}{z_0}$$

Let $w_0 = f(z_0)$ and fix $\epsilon > 0$. Since,

$$\lim_{w \to w_0} \frac{e^w - e^{w_0}}{w - w_0} = e^{w_0} = z_0$$

Since $e^{w_0} \neq 0$, then there exists $\delta' > 0$, such that,

$$0 \leq |w - w_0| < \delta' \implies e^w - e^{w_0} \neq 0$$

and

$$\left| \frac{w - w_0}{e^w - e^{w_0} - \frac{1}{z_0}} \right| < \epsilon$$

Since $f$ is continuous, $\exists \delta > 0$ such that $0 \leq |z - z_0| < \delta$, we have,

$$|f(w) - f(w_0)| < \delta'$$

Now if we write $w = f(z)$, then,

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - \frac{1}{z_0}\right|$$
$$= \left|\frac{w - w_0}{e^w - e^{w_0}} - \frac{1}{z_0}\right| < \epsilon$$

Since $0 \leq |w - w_0| < \delta'$,

$$f'(z_0) = \frac{1}{z_0}$$

□

**Definition 2.10.** *Suppose that* $u : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ *has continous 2nd partials in* $D$*. Then* $u$ *is harmonic on* $D$ *if,*

$$\Delta u(x, y) = \frac{\partial u}{\partial x^2}(x, y) + \frac{\partial u}{\partial y^2}(x, y) = 0$$

*That is,*

$$u_{xx}(x, y) + u_{yy}(x, y) = 0$$

**Proposition 2.11.** *Suppose* $D$ *is a domain and that* $f(x + iy) = u(x, y) + iv(x, y)$ *is in* $H(D)$*. If* $u$ *and* $V$ *have continuous 2nd partials, then* $u$ *and* $v$ *are harmonic in* $D$*.*

*Proof.* We can use Clairaut's Theorem.

$$u_{xx} = \frac{\partial}{\partial x}u_x = \frac{\partial}{\partial x}v_y = \frac{\partial}{\partial y}v_x = -\frac{\partial}{\partial y}u_y = u_{yy}$$

Note that $v = \Re(-if)$ thus this proof works for the imaginary part too. □

**Definition 2.12.** *If* $u$ *is harmonic in* $D$*, then we call* $v$ *a harmonic conjugate in* $D$ *if*

$$f(z) = u(z) + iv(z)$$

*is holomorphic in* $D$*.*

**Lemma 2.13.** *Suppose that* $v$ *and* $w$ *are harmonic conjugates for* $u$ *in a domain* $D$*. Then* $\exists r \in \mathbb{R}$ *such that,*

$$w(z) = v(z) + r \qquad \forall z \in D$$

*By assumption,* $f = u + iv$ *and* $g = u + iw$ *are in* $H(D)$*. Thus,*

$$h = -i(f - g) \in H(D)$$

*but this is real-valued, hence constant.*

**Exercise 2.2.12** Show that,

$$u(z) = \ln(|z|) = \frac{1}{z}\ln(x^2 + y^2)$$

is harmonic in $\mathbb{C} \setminus \{0\}$. But $u$ does not have a harmonic conjugate in $\mathbb{C} \setminus \{0\}$.

**Corollary 2.13.1.** *There is no branch of the logarithm in* $\mathbb{C} \setminus \{0\}$*.*

*Sketch.* We have $\mathrm{Log}(z) = \ln(|z|) + i\,\mathrm{Arg}(z)$ is holomorphic in $D^* = \mathbb{C} \setminus (-\infty, 0]$. We can take another branch,

$$\mathcal{L}_0(z) = \ln(|z|) + i\arg_0(z)$$

where $\arg_0 \in (0, 2\pi]$. This is holomorphic on $D_0^* = \mathbb{C} \setminus [0, \infty)$. Thus, $u(z) = \ln(|z|)$ is harmonic everywhere on $\mathbb{C} \setminus \{0\}$.

But $u$ has no harmonic conjugate on $\mathbb{C} \setminus \{0\}$. If $v$ was such a harmonic conjugate, it would be a harmonic conjugate to $u$ in $D^* = \mathbb{C} \setminus (-\infty, 0]$.

That is, $v(z) = \mathrm{Arg}(z) + r$ for all $z \in D^*$. □

## 3. Line Integrals:

**Definition 3.1.** *If $X$ is a topological space, then a curve in $X$ is a continuous function $\gamma : [a, b] \subset \mathbb{R} \to X$. We call $[a, b]$ the parameter space and write $\gamma^*$ for the image $\gamma([a, b]) \subset X$. If $\gamma(a) = \gamma(b)$, we call $\gamma$ closed and we also have $\gamma(s) \neq \gamma(t)$ when $a \leq s < t < b$, then we say $\gamma$ is a simple closed curve.*

We're almost always going to have $X \subset \mathbb{C}$. Then we write,

$$\gamma(t) = x(t) + iy(t)$$

We let

$$\gamma'(t) = \lim_{h \to 0} \frac{1}{h}(\gamma(t + h) - \gamma(t))$$

if it exists. This happens if and only if $x'(t)$ and $y'(t)$ exist. Then,

$$\gamma'(t) = x'(t) + y'(t)$$

We say that $\gamma$ is smooth if $\gamma'(t)$ exists and is non-zero for all $t \in [a, b]$.

**Example.** *Consider $\gamma_1(t) = t + it^2$ if $t \in [-1, 1]$ and $\gamma_2(t) = t^3 + it^6$ for $t \in [-1, 1]$. The first one is smooth and the second one is not.*

**Example.** *Take $\gamma(t) = t^3 + it^2$ for $t \in [-1, 1]$. This is $y = x^{\frac{2}{3}}$.*

**Theorem 3.2** (Jordan Curve Theorem.)**.** *If $\gamma$ is a Jordan curve, then $\mathbb{C} \setminus \gamma^*$ consists of two domains, each of which has $\gamma^*$ as its boundary.*

**Remark.** *Since $\gamma^*$ is compact, it is contained in some $D_R(0)$ and hence exactly one of the components of $\mathbb{C} \setminus \gamma^*$ is bounded and we call that the interior of $\gamma$.*

**Definition 3.3.** *A curve $\gamma : [a, b] \to \mathbb{C}$ is called a path or a contour if there is a partition*

$$\mathcal{P} = \{a = t_0 < t_1 < \cdots < t_n = b\}$$

*of $[a, b]$ such that $\gamma'(t)$ is continuous on $[t_{k=1}, t_k]$ for $1 \leq k \leq n$. If $D \subset \mathbb{C}$, then we say $\gamma$ is a path in $D$ if $\gamma^* \subset D$.*

**Definition 3.4.** *Let $\gamma : [a, b] \to \mathbb{C}$ be a curve. Say $\gamma(t) = x(t) + iy(t)$. We define,*

$$\int_a^b \gamma(t)dt = \int_a^b x(t)dt + i\int_a^b y(t)dt$$

*If $\gamma$ is continuous, we have,*

$$\int_a^b \gamma'(t)dt = x(b) - x(a) + i(y(b) - y(a))$$

**Remark.** *You can do the whole Riemann sums business and then use the completeness of $\mathbb{C}$ to show this, but we just skip that by defining this.*

**Proposition 3.5.** *Suppose $\gamma$ and $\sigma$ are curves defined on $[a, b]$. Then,*

*a)* $\int_a^b(\gamma(t) + \sigma(t)) = \int_a^b \gamma(t)dt + \int_a^b \sigma(t)dt$.
*b) If $\alpha \in \mathbb{C}$, then* $\int_a^b \alpha\gamma(t)dt = \alpha \int \gamma(t)dt$.
*c)* $\left|\int \gamma(t)dt\right| \leq \int |\gamma(t)|dt$

*Proof.* We are not allowed to use the word trivial. Thus, we say that the proof for a) is straightforward. Similarly, b) is routine if $\alpha \in \mathbb{R}$. If $\alpha = a + ib$, we can multiply it out and use the defintion. For c), we let $\tau$ be a unimodular constant such that,

$$|\int \gamma(t)dt| = \tau \int \gamma(t)dt$$

Now,

$$\begin{aligned}
\left|\int \gamma(t)dt\right| &= \int \tau\gamma(t)dt \\
&= \Re(\int \tau\gamma(t)dt) \\
&= \int \Re(\tau\gamma(t))dt \\
&\leq \int |\tau\gamma(t)|dt \\
&= \int |\gamma(t)|dt
\end{aligned}$$

□

**Definition 3.6.** *If $\gamma : [a, b] \to \mathbb{C}$ is a path, then the opposite path, $-\gamma$ is $-\gamma : [a, b] \to \mathbb{C}$ where*

$$-\gamma(t) = \gamma(b + a - t)$$

**Definition 3.7.** *If $z, w \in \mathbb{C}$, then $[z, w]$ is the path $\gamma : [0, 1] \to \mathbb{C}$ such that $\gamma(t) = z + t(w - z)$.*

**Remark.** *Note that $-[z, w] = [w, z]$.*

**Remark.** *If $\gamma : [a, b] \to \mathbb{C}$ is a path and $\gamma(t) = x(t) + iy(t)$ then there exists $\mathcal{P} = \{a = t_0 < t_1 < \cdots < t_n = b\}$ such that $\gamma'(t)$ is continuous on $[t_{k-1}, t_k]$. Then $x'$ and $y'$ are Riemann integrable on $[a, b]$.*

**Definition 3.8.** *If $\gamma : [a, b] \to \mathbb{C}$ is any curve, then we can define,*

$$\int_a^b \sigma(t)\gamma'(t)dt = \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} \sigma(t)\gamma'(t)dt$$

*Similarly, we'll write,*

$$\int |\gamma'(t)|dt = \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} |\gamma'(t)|dt$$

**Definition 3.9.** *If $\gamma : [a, b] \to \mathbb{C}$ is a path and if $f \in C(\gamma^*)$. Then, the path integral (or contour integral) of $f$ along $\gamma$ is,*

$$\int_\gamma f(z)dz = \int_a^b f(\gamma_)\gamma'(t)dt$$

This is really a linear combination of line integrals of. Look at Ex 3.2.6.

If $f, g \in C(\gamma^*)$, then,

$$\int (f(z) + g(z))dz = \int_\gamma f(z)dz + \int_\gamma g(z)dz \int \alpha f(z)dz = \alpha \int f(z)dz \int_{-\gamma} f(z)dz = -\int_\gamma f(z)dz$$

**Remark.** *We call $\gamma : [0, 2\pi] \to \mathbb{C}$ given by*

$$\gamma(t) = a + re^{it}$$

*the postively oriented circle of radius $r$ centered at $a$.*

**Lemma 3.10** (Homer's Cauchy Theorem)**.** *Let $\gamma(t) = a + re^{it}$ with $r > 0$ and $t \in [0, 2\pi]$. Then if $n \in \mathbb{Z}$, we have,*

$$\int_\gamma (z - a)^n dz = \begin{cases} 0 & \text{if} n \neq -1 \\ 2\pi i & n = -1 \end{cases}$$

*Proof.* If $\sigma(t) = e^{\alpha t}$ with $\alpha \in \mathbb{C}$, then,

$$\sigma'(t) = \alpha e^{\alpha} t$$

Thus, $\gamma'(t) = rie^{it}$. If $n \neq -1$, then,

$$\begin{aligned} \int_\gamma (z - a)^n dz &= \int_\gamma (re^{it})^n rie^{it} dt \\ &= r^{n+1} i \int_0^{2\pi} e^{i(n+1)t} \\ &= ir^{n+1} \frac{e^{in+1t}}{i(n+1)} \Big|_0^{2\pi} = 0 \end{aligned}$$

□

**Definition 3.11.** *Two paths $\gamma_1$ and $\gamma_2$ with $\gamma_1^* = \gamma_2^*$ are called equivalent if*

$$\int_{\gamma_1} f(z)dz = \int_{\gamma_2} f(z)dz$$

*for all $f \in C(\gamma_1^*)$.*

**Lemma 3.12** (W 3.20)**.** *Suppose $\gamma : [a, b] \to \mathbb{C}$ is a path and that $\phi : [c, d] \to [a, b]$ is differentiable with $\phi(c) = a$ and $\phi(d) = b$. Let $\sigma(t) = \gamma(\phi(t))$. Then $\sigma$ is a path defined on $[c, d]$ which is equivalent to $\gamma$.*

*Proof.* **TODO:** Do this. □

**Corollary 3.12.1.** *Every path is equivalent to a path defined on* $[0, 1]$.

Now we define the join.

**Definition 3.13.** *Suppose* $\gamma : [a, b] \to \mathbb{C}$ *and* $\sigma : [c, d] \to \mathbb{C}$ *are paths so that* $\gamma(b) = \sigma(c)$. *Then theire join is the path* $\gamma + \sigma : [0, 1] \to \mathbb{C}$ *given by,*

$$\gamma + \sigma(t) = \begin{cases} \tilde{\gamma}(t) & 0 \leq t \leq \frac{1}{2} \\ \tilde{\sigma}(t) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

*where* $\tilde{\gamma}$ *and* $\tilde{\sigma}$ *are equivalent to* $\gamma$ *and* $\delta$ *and defined on* $\left[0, \frac{1}{2}\right]$ *and* $\left[\frac{1}{2}, 1\right]$.

**Remark.** *We could be definite and write ....* ***TODO:*** *Finish this.*

**Lemma 3.14** (W 3.7)**.** *Suppose* $\gamma : [a, b] \to D$ *is a path in a domain* $D$ *and* $f \in H(D)$. *Suppose that* $f'$ *is continuous and* $\gamma'$ *is continuous. Then* $\sigma(t) = f(\gamma(t))$ *is a path on* $[a, b]$ *such that,*

$$\sigma'(t) = f'(\gamma(t))\gamma'(t)$$

*Proof.* **TODO:** Finish this. □

**Theorem 3.15** (Fundamental Theorem for path integrals)**.** *Suppose* $D$ *is a domain and that* $F : D \subset \mathbb{C} \to \mathbb{C}$ *is continouso. Suppose that* $F \in H(D)$ *and* $F'(z) = f(z)$ *for all* $z \in D$. *Then if* $\gamma : [a, b] \to D$ *is any path in* $D$,

$$\int_\gamma f(z)dz = F(\gamma(b)) - F(\gamma(a))$$

*Proof.* We have,

$$\begin{aligned} \int_\gamma f(z)dz &= \int_a^b f(\gamma(t))\gamma'(t)dt \\ &= \sum_{k=1}^n \int_{t_{k-1}}^{t_k} f(\gamma(t))\gamma'(t)dt \end{aligned}$$

where $\gamma'$ is continouos on $[t_{k-1}, t_k]$. But,

$$\frac{d}{dt}(t \mapsto F(\gamma(t))) = f(\gamma(t))\gamma'(t) = \sum_{k=1}^n F(\gamma(t_k)) - F(\gamma(t_{k-1}))$$

□

**Definition 3.16.** *Let* $\gamma : [a, b] \to \mathbb{C}$ *is a path, then the length of* $\gamma$ *is,*

$$l(\gamma) = \int_a^b |\gamma'(t)|dt$$

(See Remark 3.2.7)

**Warning.** *Length is not equal for equivalent paths.*

**Proposition 3.17.** *Suppose* $\gamma : [a, b] \to \mathbb{C}$ *is a path and* $f \in C(\gamma^*)$. *Then,*

*(1) If* $|f(z)| \leq M$ *for all* $z \in \gamma^*$*, then,*

$$|\int_\gamma f(z)dz \leq Ml(\gamma)$$

*(2) If* $(f_n) \subset C(\gamma^*)$ *and if* $f_n \to f$ *uniformly on* $\gamma^*$*, then,*

$$\lim_{n\to\infty} \int_\gamma f_n(z)dz = \int_g \mathrm{ammaf}(z)dz$$

*Proof.* For (a), we have,

$$\begin{aligned} |\int_\gamma f(z)dz| &= |\int_a^b f(\gamma(t))\gamma'(t)dt| \\ &\leq \int_a^b |f(\gamma(t))||\gamma'(t)|dt \\ &\leq M\int_a^b |\gamma'(t)|dt = Ml(\gamma) \end{aligned}$$

For b), let $\epsilon > 0$. Then $\exists N$ such that $n \geq N$ implies for all $z \in \gamma^*$,

$$|f_n(z) - f(z)| < \frac{\epsilon}{l(\gamma)+1}$$

Now using (a),

$$\begin{aligned} |\int_\gamma f(z)dz - \int_\gamma f_n(z)dt| &= |\int_\gamma (f(z) - f_n(z))dz \\ &\leq \frac{\epsilon}{l(\gamma)+1} l(\gamma) < \epsilon \end{aligned}$$

□

**Theorem 3.18** (W 3.31)**.** *Suppose that* $\gamma$ *is a path and that* $f \in C(\gamma^*)$*. Let* $D = \mathbb{C} \setminus \gamma^*$*. Define,*

$$F(z) = \int_\gamma \frac{f(w)}{w-z}dw \qquad \forall z \in D$$

*Then* $F$ *is analytic on* $D$*. For* $n \geq 0$*,*

$$F^{(n)}(z) = n!\int_\gamma \frac{f(w)}{(w-z)^{n+1}}dw$$

*Moreover, the radius of converge of the series for* $F$ *about* $a \in D$ *is at least* $\sup\{R : D_R(a) \subset D\}$*.*

*Proof.* Fix $a \in D$. Since $D$ is open, there exists $R > 0$ such that $D_R(a) \subset D$. If $z \in D_R(a)$ and $\omega \in \gamma^*$, then

$$\left|\frac{z-a}{w-a}\right| \leq \frac{|z-a|}{R} < 1$$

Hence,

$$\begin{aligned}\frac{1}{w-z} &= \frac{1}{w-a-(z-a)} \\ &= \frac{1}{w-a}\frac{1}{1-\frac{z-a}{w-a}} \\ &= \frac{1}{w-a}\sum_{n=0}^{\infty}\left(\frac{z-a}{w-a}\right)^n \\ &= \sum_{n=0}^{\infty}\frac{(z-a)^n}{(w-a)^{n+1}}\end{aligned}$$

Note that if $w \in \gamma^*$, then $|w-a| \geq R$ and $\exists M$ such that $|f(w)| \leq M$ for all $w \in \gamma^*$. Hence,

$$\left|f(w)\frac{(z-a)^n}{(w-a)^{n+1}}\right| \leq \frac{M}{R}\left(\frac{|z-a|}{R}\right)^n$$

Thus, for fixed $z \in D_R(a)$,

$$\frac{f(w)}{w-z} = \sum_{n=0}^{\infty} f(w)\frac{(z-a)^n}{(w-a)^{n+1}}$$

uniformly in $w$ by Weierstrauss $M$-test.
Thus, $\forall z \in D_R(a)$, we have,

$$\begin{aligned}F(z) &= \int_\gamma \frac{f(w)}{w-z}dw \\ &= \int_\gamma \left(\sum_{n=0}^{\infty} f(w)\frac{(z-a)^n}{(w-a)^{n+1}}\right)dw \\ &= \sum_{n=0}^{\infty}\left(\frac{f(z)}{(w-a)^{n+1}}dw\right)(z-a)^n\end{aligned}$$

Thus, $F$ is analytic and

$$\frac{F^{(n)}(a)}{n!} = \int_\gamma \frac{F(w)}{(w-a)^{n+1}}dw$$

□

**Remark.** *Recall, that if* $F(z) = \sum_{n=0}^{\infty} f_n(z)$*, converges uniformly, we can do the following,*

$$\begin{aligned}s_n \to F &\implies \int_\gamma s_n \to \int_\gamma F(z)dz \\ &\implies \sum_{k=0}^{n}\int_\gamma F_k(z)dz \to \int_\gamma F(z)dz\end{aligned}$$

## 4. Cauchy's Theorem

**Definition 4.1.** *If $\gamma$ is a closed path and $z_0 \in \mathbb{C} \setminus \gamma^*$, then the index of $\gamma$ about $z_0$ is*

$$\mathrm{Ind}_\gamma(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z - z_0} dz$$

The antiderivative of $\frac{1}{z-z_0}$ is

$$\ln(|z - z_0|) + i \arg(z - z_0)$$

Then,

$$\mathrm{Ind}_\gamma(z_0) = \frac{1}{2\pi}(\arg(\gamma(b) - z_0) - \arg(\gamma(a) - z_0))$$

That is, this counts the number of times $\gamma$ loops around $z_0$.

**Remark.** *Since $\gamma^*$ is compact, there exist $R > 0$ such that $\gamma^* \subset D_R(0)$ and since $D_R(0)^c$ is connected, exactly one o the connected components of $\mathbb{C} \setminus \gamma^*$ is unbounded.*

**Theorem 4.2.** *Let $\gamma : [a, b] \to \mathbb{C}$ be a closed path. Then $\mathrm{Ind}_\gamma$ is integer valued on $D = \mathbb{C} \setminus \gamma^*$. Furthermore, $\mathrm{Ind}_\gamma$ is constant on the connected components of $D$ and $0$ on the unbounded component.*

*Proof.* Fix $z \in D$. Then

$$2\pi i \, \mathrm{Ind}_\gamma(z) = \int_a^b \frac{\gamma'(t)}{\gamma(t) - z} dz$$

Now, let,

$$\phi(t) = \exp\left(\int_a^t \frac{\gamma'(s)}{\gamma(s) - z}\right)$$

To see that $\mathrm{Ind}_\gamma(z) \in \mathbb{Z}$, then it suffices to see that $\phi(b) = 1$. Now as in the proof of the FTFLI,

$$\phi'(t) = \frac{\gamma'(t)}{\gamma(t) - z}\phi(t)$$

whenever $\gamma'(t)$ exists.

Hence, there exists a finite set $F \subset [a, b]$ such that,

$$\phi'(t) \cdot (\gamma(t) - z) = \gamma'(t)\phi(t)$$

Therefore,

$$g(t) = \frac{\phi(t)}{\gamma(t) - z}$$

is continous in $[a, b]$ and $g'(t)$ if $t \in [a, b] \setminus F$.

This implies $g$ is constant. Since $\phi(a) = 1$,

$$\frac{\phi(t)}{\gamma(t) - z} = \frac{1}{\gamma(a) - z}$$

$$\implies \phi(t) = \frac{\gamma(t) - z}{\gamma(a) - z}$$

Last time, we saw that $\operatorname{Ind}_\gamma \in H(D)$. Hence, $\operatorname{Ind}_\gamma$ is continuous on $D$. Thus it is constant on connected components.

Next, we show it is zero on the unbounded component. Since $\gamma^*$ is compact, let $R$ be such that $\gamma^* \subset D_R(0)$. Then if $|a| > R$, then,

$$\begin{aligned} |\operatorname{Ind}_\gamma(a)| &= \frac{1}{2\pi}\left|\int_\gamma \frac{1}{z-a}dz\right| \\ &\leq \frac{1}{2\pi}\frac{1}{|a|-R}l(\gamma) \end{aligned}$$

But $\operatorname{Ind}_\gamma$ is constant on components and $|a|$ is large enough, $|\operatorname{Ind}_\gamma(a)| < 1$, thus its constant on the unbounded component and equal to $0$. □

**Lemma 4.3.** *Let $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$. Then if $z \notin \gamma^*$,*

$$\operatorname{Ind}_\gamma(z) = \begin{cases} 1 & |z-a| < r \\ 0 & |z-a| > r \end{cases}$$

*Proof.* We know $D = \mathbb{C} \setminus \gamma^*$ has two components and $\{z : |z-a| > r\}$ is the unbounded one. But $\operatorname{Ind}_\gamma(a) = 1$ by Homer's Cauchy's theorem. Then, the index has to be $1$ on the interior of the circle. □

**Remark.** *By the FTFLI, if $F$ has an antiderivative in a domain $D$ and if $\gamma$ is a closed path in $D$, then,*

$$\int_\gamma f(z)dz = 0$$

**Lemma 4.4.** *If $\gamma$ is a closed path, then,*

$$\int_\gamma z^d z = 0 \qquad n = 0, 1, \ldots$$

*and if $0 \in \gamma^*$, then,*

$$\int_\gamma z^n dz = 0 \qquad n = -2, -3, \ldots$$

**Remark.** *Recall that if $a, b, c \in \mathbb{C}$, then $\Delta(a, b, c)$ is the convex hull of $\{a, b, c\}$. By definition,*

$$\partial\Delta(a, b, c) = [a, b] + [b, c] + [c, a]$$

**Theorem 4.5** (Cauchy's Theorem for Triangles)**.** *Suppose $D$ is open and $\Delta(a, b, c) \subset D$. Suppose that $F : D \subset \mathbb{C} \to \mathbb{C}$ is continuous and $f \in H(D \setminus \{p\})$ for some $p \in D$. Then,*

$$\int_{\partial\Delta} f(z)dz = 0$$

*Proof.* We start by assuming $p \notin D$. If $a, b, c$ are colinear, then the result is easy.

Otherwise, let $a', b', c'$ be the midpoints of $[a, b]$, $[b, c]$, and $[c, a]$. We make some new triangles,

$$\begin{aligned}\Delta^1 &= \Delta(a, a', c') \\ \Delta^2 &= \Delta(a', b, b') \\ \Delta^3 &= \Delta(b', c, c') \\ \Delta^4 &= \Delta(a', b', c')\end{aligned}$$

Then,

$$J = \int_{\partial\Delta} f(z)dz = \sum_{k=1}^{4} \int_{\partial\Delta^k} f(z)dz$$

For some $k$,

$$\left|\int_{\partial\Delta^k} f(z)dz\right| \geq \frac{1}{4}|J|$$

Call that triangle $\Delta_1$. Then $\Delta_1$ is similar to $\Delta$ with sides $\frac{1}{2}$ the length of the original. Now, iterate to get,

$$\Delta \supset \Delta_1 \supset \Delta_2 \supset \dots$$

Next, note that,

(1) $l(\partial\Delta_n) = 2^{-n}L$ where $L = l(\partial\Delta)$.
(2) $|J| \leq 4^n \left|\int_{\partial\Delta_n} f(z)dz\right|$
(3) $\operatorname{diam}(\Delta_n) \leq 2^{-n}L$.

Since $\Delta$ is compact, $\Delta_n$ is closed in $\Delta$, and the diameters go to $0$, there exists a unique $w \in \Delta$ such that $w \in \Delta_n$.

Since $p \notin \Delta$, $f'(w)$ exists. Thus, given $\epsilon > 0$, there exists $\delta$ such that if $|z - w| < \delta$ then,

$$|f(z) - f(w) - f'(w)(z - w)| < \epsilon|z - w|$$

We can assume $n$ is large enough such that $z \in \Delta_n \implies |z - w| < \delta$.

Hence,

$$|f(z) - f(w) - f'(w)(z - w)| < \epsilon|z - w| < \epsilon 2^{-n}L$$

On the other hand, by a previous lemma (FLFLI),

$$\int_{\partial\Delta_n} = \int_{d\Delta_n} (f(z) - f(w) - f'(w)(z - w))dz$$

Thus,

$$\begin{aligned}\left|\int_{\partial\Delta_n} f(z)dz\right| &\leq \epsilon 2^{-n}Ll(\partial\Delta_n) \\ &\leq \epsilon(2^{-n}L^2)..\end{aligned}$$

Now by $(b)$,

$$\left|\int_{\partial\Delta} f(z)dz\right| = |J| \leq \epsilon L^2$$

Thus, taken care of the case $p \notin \Delta$. Now suppose $p$ is vertex. Say $p = a$. Since $f$ is continuous on the compact set $\Delta$, there exist $w \in [a, b]^*$ and $y \in [c, a]^*$, such that,

$$\left|\int_{\partial\Delta(a,w,y)}\right| < \epsilon$$

Then, by inspecting the picture, we get,

$$J = \int_{\partial\Delta(a,w,y)} f(z)dz + \int_{\partial\Delta(w,b,c)} f(z)dz + \int_{\partial\Delta(w,c,y)} f(z)dz$$

The latter two are $0$, thus, $|J| < \epsilon$ and $J = 0$ as $\epsilon$ is arbitrary. $\square$

In the next lecture, we will upgrade this.